

ActInf Livestream #029.1 ~ "Active Inferants: An Active Inference Framework"

Livestream | 1:51:53

looks good go for it

hello everyone welcome to the active

inference lab live stream number 29.1

today is september 21st 2021 and we are

going to be discussing this paper active

inference and active inference framework

for ant colony behavior

this is a recorded and an archived live

stream so please provide us with

feedback so we can improve our work we

are a participatory online lab that is

communicating learning and practicing

applied active inference you can find us

at all these social media links

and all backgrounds and perspectives are

welcome here we will be following good

video etiquette or livestream

here at the short link you can find all

of our upcoming live streams that are

planned

um and if you want to get in touch or if

you want to participate please reach out

to us and we will invite you to

participate in a panel

today our goal is to learn and discuss

this paper active inference an active

inference framework for ant colony

behavior

uh we will be maybe going through some

of the aims and claims keywords figures

uh and

having a discussion around this paper

so just to start off we will

go around and introduce ourselves and

then start off with some warm-up

questions so my name is blue knight i am
an independent research consultant from
new mexico and i will pass it to marco
so hi

my name is marco i'm located in vienna
austria i'm a postdoctoral fellow at the
conrad lawrence institute for commission
research

and my interest is very much from the
clinical perspective i have to do with
dbs patients and i have come across
active inference actually rather
recently

cool daniel

thanks um i'm daniel i'm in california
and
as

first author i'm happy to maybe just
introduce the
big picture

um you wanna just briefly say hello and
then i'll return

stephen are you

here steven

yeah hi there

so did you just introduce me hi stephen

here um nice to be on and uh

i am based in toronto and um

i work

with communities and work with spatial
sense making

around different issues and um how
active inference can now

help to understand

those processes of sense making

and

dynamic inference in in the modern world

awesome so daniel do you want to maybe

introduce the paper or give us a summary
of what was your big takeaway message uh
from this paper or what was your
motivation behind the work
well just
hearing some of the backgrounds of those
here ranging from human medical and
clinical to sort of social and
collective processes playing out in
humans
it just shows that we each approach
active inference from our own experience
so
personally my background was in genetics
and insect behavioral ecology
and so it was conversations and reading
the papers
of especially axel and of course carl
fristen
and we were having conversations about
where does active inference fit into
discussions about collective
intelligence where does active inference
fit into bigger discussions about
evolutionary theory like if active
inference can be an integrative
framework
for different social systems for
different human systems
why not other animals too
and uh so there we were at the
intersection of
curious about big questions in
evolutionary theory which i know we'll
return to today and specific questions
about the areas that i was researching
at the time
which was the regulation of foraging and

decision making in ants and so it was conversations with with axel and maxwell to a large extent where we started thinking how could we modify the types of models that we currently have in active inference and bring that into something that adequately at least starts to work towards capturing how the ants do it and alec had worked on the code of many active inference projects and so that was quite a fun process maybe we can also talk about like interfacing between ants and computer science and we mainly wanted to navigate a little bit towards the bigger questions in biology but do so by really tackling a specific scenario that would then lend itself to unique explanations and predictions and models and the specific contribution that differentiates the active inference model from other active inference models is first that it's in insects and ants but also importantly that it's stigmatic so it's modeling explicitly the feedback between the environments and the agents whereas there had been sort of multi-agent active inference simulations where the agents communicated with each other or they had a shared theory of mind those types of models existed but this was the first time that one of the action affordances of the agents

included the capacity to modify the environment and the status of the environment modified the likelihood of the agent taking different kinds of actions

cool so i just want to reach out to um marco and steven and just ask like what's something that you were excited to talk about today or something that you liked or remembered about the paper either you have thoughts should i start

so one thing as i already mentioned before is the role of dopamine which i find super interesting and you mentioned the paper that the foragers have higher levels of dopamine so i was wondering about the role of that and how that influences

affordances and what the individual and also the level above like the the colony does with that so is there a specific role for dopamine

for the for the colony as a whole great question and we wrote it down on slide 14 so we'll definitely return to talking about some of the molecular roles of dopamine

cool and stephen do you have any thoughts

yeah i suppose mine's a bit more of a broad question but i'll be really curious about that balance between sort of specific

um things relating to ants and how that relates to things like dopamine and

maybe the broader process of like the
design thinking and working through this
kind of active inference framework just
how that was useful
to think about
how things may relate to their
environment
so i suppose the process of creating the
model as well as what the model was and
what it found out about things in the
world thanks
awesome daniel
it's a great point um almost asking
about how do we study the biomimicry not
of just materials like light materials
or strong materials but of collective
algorithms
and we can't really study
the success or the attributes of a
collective behavioral algorithm outside
of its ecological niche
and so
a very long and important branch of
research by my phd advisor professor
deborah gordon which is where i was when
we were doing a lot of this initial work
she studied for 30 plus years
understanding how the statistical
regularities of the niche
were then
associated with which collective
behavioral algorithms of ants
we saw happening
and so there's like 15 000 ant species
there on every piece of land except for
basically antarctica which is kind of
where they should be if you think about
it but

if you want to have a framework for ant behavior that includes the desert and the jungle and all these types of different environments it sort of baked in

this idea of appropriate generalization because like we definitely can't have a theory for each ant species so we need to look a little bit bigger than that yet we don't want to have just an ant specific theory only and let's you know think about termites and bees and non-insects so what is the appropriate level of generalization and how are we going to generalize the study of collective behavioral algorithms and how they interface with different kinds of environments

while also remembering that you know we got to have our eyes on the ground we need to be explaining those ants we can't just simply spiral off into the abstractions about collective behavior so they answer a cool system because they remind us about ecological and behavioral diversity and the need to generalize but also there is a real system that we can actually see awesome thanks

steven

yeah thanks daniel i was wondering how much um that generalizability that is in the field that you're in with your supervisor you know that might be something that really brings something to the table for active inference and i'll be curious as well

you know how much those algorithms that

have been worked on for a number of years
either got modified or ported or had to be kind of reconceptualized from the ground up you know
that'd be interesting
cool daniel you have a response
yep so one reason why active inference is able to swap in to where other theories or frameworks have been used before is because at the core it's an input output structure just like a lot of other behavioral studies like sensory information comes in
something happens
and then action comes out so in that way it's able to map onto a lot of other ways that people have thought about animal behavior or ethology
one of the differences in active inference is that we're not committed to any specific cognitive model so like in this paper we just used pretty much the most simple possible cognitive model if you could even call it that it was one step decision making there was no memory there was no anticipation there was no capacity to use some of the sensory affordances that we know that ants actually have and use in nature like multiple sense interactions with nest mates interactions with other physical objects so we mapped onto this input output structure
but we know that we can generalize it later but so that's one reason why active inference will um hopefully play a nice generalizing role there

um

i i hope yeah go ahead stephen yeah
that's really cool and i suppose one
thing that when you mention that i
suppose this even ties in with that
dopamine question but i'm not sure
because i'm not as familiar with that
but

is by not having it all based on the
cognitive it's showing what the niche
can do for you in a way it's like how
much

you know we all assume it's all going on
in the head which is the old way of
thinking but

with the active influence it's like well
maybe there's other ways that we can get
where we need to get so to speak
awesome

and with the ants it was almost never a
question of it happening inside the head
only because it's it's not to demean
their cognition it's just very clear
that no single forager would know how
many seeds does the colony have how many
does it need through time what's the
weather and the trade-offs of foraging
where are the seeds i mean those are
questions that at a first pass we know
are just unknowable and so
it's pretty clear that the way that
individual nest mates are making
decisions
has to have
a radically different mode
than for example let's wait till we have
all the information and then make the
well-ration decision based on our

simulation it's just not happening and
then once you see that lens on ants you
go

huh well maybe there's some other
forming settings where the agents don't
have total information and even if they
did it wouldn't be obvious what to do
like we can all imagine a chess board we
have total information about where the
pieces are but even experts in chess
don't necessarily know the best move and
there might not even be necessarily one
perfect best move

so it really moves us into a setting of
thinking about how action is undertaken
including action that can be epistemic
like gaining knowledge amidst
uncertainty

rather than waiting until we've resolved
our uncertainty because then what we
still wouldn't know what to do

cool that leads in pretty nicely to one
of the questions that i had brought up
last week so there are these colony
level metrics right like how many seeds
are

in the nest or how many you know uh
nurse nurse maids are there relative to
infants or foragers relative to you know
a baby yeah i guess

so how are these colony level metrics
retained in a colony

good

question well

you asked a why

question in a sense so it makes me think
about the tin bergens four wise which
we've spoken about um on other live

streams but the evolutionary
um why is just colonies that have
certain colony level attributes are more
likely to succeed and so we end up
seeing

the emergence of certain
ratios of male to female nest mates or
certain ratios of forage or to nurse
performance just by virtue of their
survival so that's sort of an
evolutionary circularity just like in
active inference we talk about how the
systems are resisting dissipation so
systems that are doing it it sort of
explains itself they're here and we
observe them because they're resisting
dissipation

and

it's really this question about
colony-level phenotypes that
fascinates people who study collective
behavior because there's the traits that
a nest mate has which are not so
different than the traits that any other
insect has like six legs what's the
width of the head what's the gene
expression of the gland or the mid gut
like what digestive enzymes are they
making those are things you can measure
about one nest mate and so the body plan
in ants is really quite similar to all
insects

but then there's this really
qualitatively different level of
organization with a colony
and how do those phenotypes arise
how are those colony level phenotypes in
feedback with nest mate how does

selection

shape those phenotypes

nice thanks for your answer

um and there's a couple so maybe let's

go to the um

this slide

that shows like where you're measuring

the colony level

metrics or or

the

the coherence right like the colony

coherence or i forgot the term that was

used um but you

have the model or maybe maybe we should

start with the model really i guess so

so maybe do you want to just give us

like a quick overview

of like the model and the point of it

and um yep

let's go to slide 15 and so just to

bridge from the colony phenotypes back

to the specifics of the model it's sort

of like if there was a music festival

there's all kinds of phenotypes of the

individual participants like their

self-report where did they move how is

their decision-making being influenced

by their sensory stimuli but then there

would be certain descriptions

that you'd only have

one of for the whole music festival like

what is the average distance of people

so that's what we were studying with the

swarm coherence metric

but it just points that there's going to

be two

totally different types of phenotypes

some can be measured multiple times

within a colony per nest mate and other times there's going to be phenotypes that are only arising at the level of the colony like the sex ratio or the average distance and what our model did was we specified the process by which the nest mates basically perceived and acted so it was an action and perception model at the nest mate level we didn't explicitly have a input or an output for the quality not that it couldn't be done you could also think about where that might come into play maybe in dot 2. but we looked at a setting that has been used to study colony decision making and then we specified a nest mate level model and put it in this context where people had already been using it to study the dynamics of how colonies make decisions so here it's called the t maze test and there's two arms to the tmas the two um red arrows on the left side here of figure 1a and food is on one side of the tea maze and it can either be static or it can be switching and this is something that they use in lab rat experiments as well so the colonies task is to basically get food but what we did is we didn't actually encode

that as like a colony level goal
we really went down to the nest mate
level and we specified
just like any other active inference
model what are the sensory observations
made by nest mates which in this case
was just the pheromone density of the
squares around them
what are the action affordances of each
nest mate which is sort of it can move
in any of the adjacent squares
and then
what's the generative model
which entails a preferences and a free
energy minimization happening inside the
nest mate at each time point
so that given the observations at a
certain time
the actions are selected and in this
case the preference was for more of the
pheromone so like the pheromone can be
interpreted as a
attractant something that the agent
pursues or prefers to observe
and um
the only other tweak which is what
introduced the pheromone in the first
place and the whole concept of stigma g
was that on the way in after they've
touched the seed but not before
they also deposit pheromone so that was
an additional action affordance to
deploy pheromone and there wasn't any
decision making about that it was just
if you've already touched the seed then
you lay down the pheromone and if not
you don't
and so it's sort of like those simple

rules

pursue regions of higher pheromone

density

lay down pheromone if you've touched a

seed but not if you haven't

those two rules

at the nest mate level recapitulate

some of the phenotypic dynamics of

colony decision making so that's why

it's fun to think about like maybe we

can look at complex collective phenomena

and then understand how they emerge from

simple interactions among agents and

agents in their environment

nice stephen

just a quick question is i'd be

interested because you've got say one

ant that goes off

foraging type of thing and then maybe

you've got the second one that follows

and then i suppose there's

what goes back to

the nest if i'm right um

yeah i was just wondering like

how much

more complex or does it help to have

more in the modeling or does it make it

more challenging to start using too many

ants

so we templated the natural history if

you will of this off of a formica ant

species um like a wood ant so it's

referenced in the paper

this dynamic of like go out without

leaving pheromone come back with

pheromone so given that natural history

to change the number of ants from two to

two thousand to two million it's just a

number in the config file

when you run the code so it actually is
not challenging to scale the colony's
size

it might burden your computer or cause
some other limitations and it certainly
will change the collective outcomes
maybe in a linear or maybe in a
non-linear way like we explored in the
figure three

but

there are so many natural histories and
ways that ants forage

like there's tandem

running or like an ant remembers where
the food is

and then it does like a leap frog and it
takes an uninitiated ants nest mate
and leaves it and then when those two
return they bring back four and then
there's eight so there's all kinds of
recruitment and food

localization strategies in ants which is
why it's important that there's not just
like one single way that ants forage
because they're looking for things that
have different regularities in the
environment

so

within a natural history the script is
written in a way where it's pretty easy
to change the map shape or to change the
number of ants or their appearance
dynamics

and then a little bit of the challenge
would be like to adapt it or transpose
it to a different natural history or to
add multiple interacting pheromones or

other kinds of things like interactions

amongst the ants

marco

yeah

sorry i just wanted to ask about the the

the model as it is described in the

paper because um

at the beginning the ants have basically

random movements right because all the

surrounding step

movement opportunities are the same uh

probability right

so is it

sort of like a brownian motion until the

first ant discovers the

the food source and then starts altering

the environment by

the pheromone

placement

that's exactly right and there's seven

supplemental videos gifts

where it's shown

and what it kind of looks like on this

team ace

it looks like you see these little

particles that start swelling and

diffusing up from the bottom so that's a

total brownian walk because there's no

preference for direction when there's no

pheromone deposited

and then

you see one ant find the seed

and then it starts laying down this

pheromone but it's also diffusing

randomly

because at least initially there's also

no preference for direction but then you

can imagine that there starts being a

higher concentration of pheromone on the side with food
than not and eventually that gets reinforced because it starts to attract anyone who sort of diffuses into this regime
of a locally increased pheromone density they start to get pulled to the places where there's increased density and then that ends up reinforcing and as soon as there's like a ridge of high pheromone connecting the food back to the nest entrance
that's the formation of a pheromone trail so no ant needed a bird's eye view or a goal to even form a trail
it's a purely agent-level process-based explanation
by which you get the emergence of a phenotype that goes beyond but one nest mate knows or even prefers and that's the existence of a trail now you can imagine that colonies that have the appropriate parameters like the right amount of explore versus exploit how much to follow one pheromone how rapidly should the pheromone decay those parameters are actually shaped by the evolutionary process
colonies with maladaptive decision-making of nest mates are not going to bring home the seeds they're not going to reproduce
whereas for any given niche if the regularities can be captured by the decision-making algorithm that the ants use
you'll see those parameters plus slight

variations on them
in the future
awesome thank you and that kind of
brings us to a figure two which i um
threw up here and
i i clicked this quote from the paper so
it's talking about the rate of pheromone
decay and it says informationally this
decay rate can be thought of
thoughts of an environmental
perturbation or drift that counteracts
the reinforcing effect of stigmatic
convergence
and so in this and
here even in the figure legend it says
like we chose the representative
simulation did not turn the parameters
to discover optimal decision making
um and so i was just wondering if you
played with the decay rate at all
because i think about like if the
pheromone didn't decay
there would be a really difficult time
switching like from one arm to the maze
to the other when the food source
switches so you have to forget at some
point where the food was then this kind
of brings me to one of the big questions
is like how important is this decay rate
in like
the memory of anthony like so this is
clearly like an example of niche
modification where they lay down the
pheromone and it externalizes their
memory of where the food is for other
ants in the colony but there's this um
you know it has to decay otherwise
they're just confused like if we were

flooded all of the time with all of our
memories like unable to distinguish our
our memory from the present time which
happens in some forms of you know uh
neuro neuro like diseases narrow
atypical people have this like you know
inability to distinguish the present
from the past from the future so
just is that like a critical thing or do
you have any comments on that maybe or
maybe stephen wants to comment on what i
said and then uh you comment or stephen
do you want to go first or what do you
think okay yeah i'll just add one thing
to that um
yeah i think that's a
um
a good point is i suppose how much does
the ability to
um
allow the environment
um
to re
i suppose re-remember i'm not really
remember but reformulate so you may not
need to forget as much as just
change how much you're weighting what
you
you um
give priority to so in a way
it's like by knowing something that
you're more certain about it might get
something similar so i'll be interested
about
that idea of
you know offloading onto the environment
and how that maybe
negate some of that anyway

yes so it is um akin to forgetting
but what's really interesting is it's
the colony for getting because the ant
doesn't have a direct memory
so as we encoded it so it's a lot like
cumulative culture
there's a totally
qualitative difference between what an
individual remembers on board and then
what the culture remembers and so we do
modification on our cultural niche
breeding scott
and so that enables new dynamics of
learning and forgetting
which are distinct have different
mechanisms than the ones that happen at
the nest slash individual person
level
and so it is um a forgetting that occurs
and otherwise you would get stigmatic
reinforcement stigmatic tightening and
it would end up just being over fit to
the way things were
and then you wouldn't be able to access
new resources or have the flexibility to
find
the seed after it moves to the other
branch
awesome
sorry did you have something else to add
how goes it scott
hi scott
um so here we'll switch on to oh steven
go ahead soon
yeah just one thing daniel i was
wondering how much you think i mean i
know it's an ant model but you know
you're talking about extrapolating to

beyond

how like you just mentioned that with
culture so that

um i wonder if there's some principles
that starting to emerge that you think
you're kind of curious about exploring
further because of that ability to
think differently about how we
know

the precursor model

to active inference was actually
on slide 53

the acquisition of culturally patterned
attention styles under active inference
so this was a visual scanning
visual foraging model where the eyes
scanned over pottery designs
and there were cultural preferences for
certain local motifs
and so

it's kind of interesting that we almost
stripped

the cultural interpretation away
to apply it to ants and i do think that
it could be

re-integrated in

it just helps us clarify that we can
know what the individual level
mechanisms are like of personal
preference what is being sensed by the
person and what are their action
affordances kind of like all the
variables in the active inference model
perception cognition and action
and then the external states
can be other agents and we've talked
about communicating agents before but
the external states can also be

something that's cumulatively modified
and so when your sensory capacities
include the ability to read websites
and then your action affordances include
the ability to modify websites
it would take transposing the model into
those different

inputs generative models and outputs
but it does suggest that we don't have
to make everything only existing as a
function of interactions amongst active
inference agents in fact cumulative
culture helps us distinguish pieces that
are clearly cognitive clearly on board
from outcomes that are clearly
cumulative and have memory that isn't
just what the agent remembers
almost gives a a rationale for fashion
in a way could be interested in as well
when you think about design and of
pottery there's a fashion that there's
there's a fashion of other things as
well

that that makes me think about like who
controls how the seed switches arms so
here it's the spatial movement of the
seed from one side to the other and the
switching dynamics but fashion i mean i
guess what people wear or just like oh
it's a fashionable book

it's sort of a way in which

people's

niche modifications include locating and
moving or hiding seeds and it's like oh
you didn't know about this band like i
foraged better because i know this band
that you don't know that's sort of like
hipster informational foraging

and so how do people signal
through fashions you know keep up with
the time cubism is the art that we all
look at this is what we think is cool or
this seed location is no longer in vogue
so maybe some types of foraging
metaphors
could help us understand cultural
phenomena
that also feels like hope
that also feels like irving goffman kind
of reference of projecting out as you
want to be perceived that those
recursivity
so that you you know if you want to be
perceived as a mountain climber you wear
a shirt from a mountain climbing store
um and so you're you're expressing a
thing and you want to be perceived as
that so there's that kind of that mirror
effect that goes on um uh
to define ourselves by how others see us
it also reminds me the earlier notion of
the cumulative embodiment
reminds me of the david bone work that
all reality is first a thought
and so the realities that are
accumulated are just accumulated
thoughts
that then become artifacts of that
interaction and then you know their
embodiment is left in that
intergenerational
communication
through language and through material
culture is where the mind resides
really in the interaction with those
things

so we're able to communicate in deep
time that way they it reminds me that
quote that
bach and mozart didn't die they turned
into their music
right so we still communicate with them
we still hear them
at who they what they were communicating
out the realities that they were trying
to create became embodied
and they still available to us through
deep time and over space as well and so
that
maybe part of the cultural affordances
are the leveraging of the communication
the leveraging of self and identity
and you start to have people able to
pick up on those and and leverage
themselves and then they take it and
take it in a slightly different
direction then you have that
innovation
that happens and one of the things i
find fascinating is in fashion and in
food are two places you do not have
intellectual property protection of
copyright but they're the most
innovative or if you're from england in
innovative areas
which is fascinating right because they
don't need the excuse for intellectual
property laws the rationale is that it
helps to promote innovation but in fact
the most innovative things don't aren't
afforded that protection they're more in
a
foraging kind of um enthusiasms that
make those things

you end up
with the intergenerational maintenance
not just construction and enshrinement
of a pyramid but the actual ongoing
maintenance of
cumulative culture
whereas
it's not enough to just do it once and
then leave it behind and so it is
interesting about how
the
process and carrying the process forward
gives you the static objects of
cumulative culture but the cumulative
culture and the static objects
themselves
they don't map one to one onto a process
let alone adaptive one or a successful
one and so how do we
and that relates to um
the way that mod that reward was modeled
in this um simulation and also to
dopamine so i know we'll get there
awesome uh stephen and then scott
this reminds me as well i was that there
was a marketing manager that i once
spoke to and he was telling me what they
they pay
people to go into bars
and to drink certain drinks in trendy
clubs in like the big cities say in
london or whatever
so that they would be seen to be
drinking that drink
and then it starts to spark off of
course it's completely covert
but this sort of thing you can see how
that distorts people's um

stigmatic trail

because they see someone who's

presumably very good looking or

fashionable or very

high status or whatever

at the bar sitting on the stool drinking

the drink in a conspicuous way and they

they take the cues at some level so

anyway i thought that's interesting

for sure

okay so i think we can maybe move into

the

oh sorry oh sorry scott

i was looking i was looking for a shoot

can you hear me

yeah

okay sorry i was looking for the

controls i totally lost my screen to

sort of screwed up um the end in the u.s

stephen there's a thing i was looking

for the word for it but there's actually

when you go into a bar and say i'd like

an absolute vodka please if you've been

hired by absolute you're required to say

i've been hired by absolute vodka i'd

like an absolute vodka please they

actually have a disclosure requirement

for that exact behavior where people go

in bars

and if you're being paid you need to

make a disclosure as a technical matter

under

under federal law so i thought that was

pretty interesting

um there was another point i was going

to make but i spaced on it because your

point i'd like your point so much

sorry

it's like an active um disclosure here's
my preferences here's my expectations
here's some components of my generative
model right
awesome
yeah i was just wondering since uh since
you were talking so much about the
distributed cognition now and making
sort of
extrapolations there
there clearly must be some limitations
there right because
i i
assume or guess that
like human culture societies
have a larger capacity to change
phenotype compared to let's say
the limited
variables that
an ant colony might have and that might
be subject to uh evolutionary pressures
and and then is there let's say is there
is there a
larger um
flexibility and phenotype for for human
culture would you say
we
almost have affordances that include
changing the hyper parameters the higher
order parameters on our
niche and clearly the scope of our niche
modification
has expanded as well
so it'd be like if this ant colony you
know we added another level it can
decide where to move the seed to or it
can decide to
modify the structure of the maze i mean

we know that ants build they build their
nest architecture in many cases so
what happens when you enable a new kind
of performance
it reveals like a whole new
game theoretic or signaling theoretic
dynamic
that can
flip
how it would have been otherwise like in
a fixed maze maybe there's a strategy
that works
but maybe if you add this ability to
move blocks
maybe the strategy is actually to um you
know make a hypotenuse just going
straight to the food so that totally
changes which parameters of learning and
forgetting
you might want to have
and so it's a very interesting question
about how
affordances come into play
how do they interact
and then
what is happening with humans with our
new affordances
our ability to
have long-range synchronous
conversations and to make irreversible
ecological decisions as well
those affordances for niche modification
and communication are changing what it
means to be human
definitely agree with that um as we kind
of mentioned in the dot
zero contextualization video
any more thoughts on that or we can

maybe move on to these colony
level metrics in figure three
figure three just to describe the two on
23.
um on the left
is just a tally of basically the seeds
returned home just the count of the
round trips
and then
that is being shown through time steps
on the x-axis
for four different numbers of
foragers like 10 30 50 and 70.
and so
again we weren't like tuning parameters
or making statistical claims but it's
interesting to see that
the 30 50 and 70 the ones that kind of
look like step funk same accumulation
dynamics as the others do
and then on the right side is that swarm
coherence the um the
distance metric amongst the nest mates
and so again
maybe this could be run out or it would
be different on the map shape and all
these other things but just to see that
the average distance sort of increases
and then stabilizes at a somewhat
characteristic rate so these are
phenotypes that nest mates don't have
but they're composed of the phenotype
the position the behavior etc of the
nest mates and so again it just points
to the need to be clear like when are we
talking about
individual level phenotypes that might
be in feedback with environments or with

collective outcomes and when are we
actually talking about specifically
collective outcomes
nice

so i have a question that um
you know relates to this directly so in
this kind of colony cohesion or i think
you call it form coherence
um there's the
distance metric that is measured but but
in real ant colonies like are there
other metrics that kind of are used to
judge this kind of cohesion because i
wonder about like the time that's spent
communicating like ant communication
time like does this also like enhance
colony performance or what other kinds
of metrics can you like use to determine
this kind of cohesion

we talked about this a lot with the
authors and we so we thought okay well
there's like the euclidean distance the
average distance or the distribution of
distances amongst estimates one could
also look at the correlation
of their trajectories so a perfect
correlation of the trajectories would be
like every ant is taking the same trip
whereas no correlation of trajectory
would mean like every ant was going on a
different direction or having different
movement dynamics and so that could be
computed over the total trajectory of
each ant or even at each time step
like on the next time step
zero correlation means like that
there's 50 going up 50 going left 50
going down 50 to the right

or maybe there's like this sort of
shift a coordinated shift within a time
step as well as among time steps so
looking at the correlation in
trajectories the mutual information
um these kinds of things could be
calculated for the colony and then we
thought you know it might not even be
easy to rank these because what if the
mutual information is low
because they're doing different types of
foraging and that is constituting a more
coherent swarm
so just because their correlation is
high
that's sort of like first pass oh well
it must be a coherent swarm because
everyone's taking the same trajectory
but then there's this other type of
coherence
that might even be more emergence
when the correlation or the mutual
information of trajectories is low
and it's actually by virtue of that that
the colonies able to achieve things that
the nest mates don't
awesome thank you

um

stephen yeah can i just ask when you
mention that

could that um about the the different
types of

um

movements and correlation

um do you think that could

point towards um

how there could be different types of
counter factuals brought to the table so

as maybe the information coming back as
well as being a seed it could be a
counterfactual um
i i suppose the war it's not a seed but
it's a wall or
um that i don't know maybe that doesn't
mean i haven't thought it through but
i'm just wondering how counterfactuals
might play out
founder of factuality is it's an
interesting question
because it's a it's a mental ascription
which
we think we can do but then
do ants do counterfactuals or how do
they at least act as if
and
it's a great question just one thought
would be
the resilience of their algorithm
is such that they act as if they're
taking counterfactuals into account
so for example their trail is functional
when it's functional
but then it's always like this adjacent
possible that the trail could be
disrupted and when the trail is
disrupted
they find a way to go around or to
repair the trail and so it's like by
having this
counterfactual like the adjacent
possible that they might not have a
functional pheromone density
there's this self-repairing and
regenerating aspect to their behavior in
nature
that makes it look as if they are

accommodating these counterfactuals and
it reminds me of one of the greatest
disinfo scenes of all time in a bug's
life i believe or maybe ants like where
leaf drifts down
and then all the ants are carrying the
seed of course in their hands not in
their mandibles which not how ants carry
seeds and then like the overseer comes
over and blows the whistle everybody
stop
you know the first ant that was
returning home gets anxious
it's like you couldn't stack
i know it's just a movie so i know chill
out etc but you couldn't stack more
misconceptions
into that representation of how
resilience actually arises in foraging
in overseer doesn't come
and the first ant
engages with the barrier
and then goes around it or over it and
that is what ends up allowing the rest
of the ants who don't have communication
with any overseer or even that first ant
to participate in that
re-functionalization of that part of the
trail so
it would be cool to see
representations that don't disinform as
to how ants actually
work
that's really cool
that's a really cool response actually i
also suppose
you could flip that as well and say well
maybe the adjacent possibles of these

paths that the ants are taking in the
environment is maybe that's how we get
our counterfactuals in the brain
it could be like
a foraging path you know
um i'm not saying that's how it is but
maybe at times there's something
that mirrors that but in in our own
cognition
scott
you know that
daniel that reminds me
of our misunderstanding of how ants work
it reminds me of the assertion i read
that dogs
in the wild don't have the alpha dog
kind of characteristic and the impact
mentality i don't know i don't know the
research on this but
that they learn that from humans
that and so it's interesting we kind of
project onto animals
human ideas about
stacking or
whatever
social status
and it's fascinating to think
whether in art because we were talking
about fashion before
might the arts
be a vehicle for
inviting humans to start to
understand things that ants know
right you know what what can we find
from the art world a movie or
something like that which did a
different treatment of it and started
children embracing that right because

these are children are learning from
these movies
it's a fascinating notion to think about
how
a more
i use the word accurate portrayal but a
more maybe sensitive portrayal to the
dynamics of other biological beings
might help us as biological beings to
find our way
because certainly the abstractions that
we've we're teaching children now are
not fully functional
this so there may there's some
additional learnings there the
conversation i was on just before this
we were talking about
um capacity building in precarious
populations and i was pointing out a
similar kind of thing we were saying
well we might have things to learn from
precarious populations it might not just
be parachuting in colonial type
solutions but rather what can we learn
from
new discoveries
and this was in the context of um the
idea of bringing technology to for
information to precarious populations
but in the reverse of it bringing
community
knowledge to technologies
the reverse flow and again similarly
here maybe the community knowledge of
ants
needs to be brought to the technology
architectures
that we're talking about rather than us

constantly being blinded to the architectural elements that we might have because we have these centralized elements these alpha dog notions these things that we project out so we're unable to learn because we're just imposing a structure on something that may not be there so we can't learn

nice daniel

the history of psychological projections onto the ants and the bees is tremendous it relates to sex and gender and politics i mean she's called the queen and she's called a she but she's just a diploid insect so how do we go from the words that we use and the kinds of models that we use to describe how the colony functions is it an economic maximizer is that the goal is gdp maximization or is getting seeds the goal and that was one of professor gordon's insights as well is like it's not just about maximizing the per capita return on foraging in terms of seeds which optimal foraging theory often suggested because in the desert it's costly to go out foraging you are spending water to get seeds which give you energy and water but it's kind of like there's a time where you don't actually need to get more seeds so the way that the projections occur and that they are mapped sometimes implicitly or explicitly to our human systems and ideologies is very interesting and i think that relates more to this general point in active

inference which is actually that sensory observations are ambiguous that's the insight of helmholtz and unconscious perception which is like the perception of ants even if the edge detection is relatively unambiguous what it is that the ants are doing whether they're taking care of every nest me and it's like utopia or they're sloughing off the useless ones and it's like hellish that's our projection and so that's why it's so interesting to have a model where we can be clear about what the sensory observations are of ants of ourselves and then how our generative model is actually what is generating perception you know we see with our mind not with our eyes so the ant is also perceiving with its mind not just its antenna awesome stephen that's a really very very good point and also i think with helmholtz bringing that back in is really useful in terms of how we project because he was a way for us to get back to what we already knew it's like with all our intelligence we as psychology came in and this knowing and this perspective that knows and that becoming this almost unspoken axiom that we didn't even know was unspoken everything from there on in was about what do you think about this

and what do you think about that and
what do you think about the other and
100 years later
you know we're in all sorts of problems
right
um well were very quickly in all sorts
of problems actually at the beginning so
suddenly yet you know going back before
then i'd be interesting to know about
300 years ago they probably look at us
with a wry smile and they'd be like
yeah right because they hadn't had a
whole load of other stuff
that
involves this knowledge of knowing as
being about thinking and us and it was
much more in the kind of
you know unknown states of the world i
mean their dictionaries weren't even
there right so they didn't have that
defined language in the same way
um so yeah i think that's that's a nice
point
well lang
language becomes oh sorry
scott and then daniel
language becomes that artifact again
that's why i keep uh yeah again it's not
it's not accurate what i say when i say
the mind resides in language and
material culture but it's i just keep
saying it to get people off thinking
that it resides in the brain
and so it's it's just
the
that
those traces we leave of prior
interactions that seem functional

become

so ascendant you're born feral

and then you kind of look and observe

what works

and it's so localized

knowledge is so localized and now it's

become the access to broad knowledge is

is so

great

that there's nothing that suggests that

the human lifespan was designed to be

able to absorb all the knowledge that's

now available right there's no we're not

necessarily

able to do it but we try i certainly try

i got i'm surrounded look at all the all

of us that have books around us you know

we're like we're surrounded by so we're

trying to

figure all this stuff out but there's

nothing that says we should be able to

and certainly not as individuals right

so then you then i start thinking well

okay i'll take solace in the fact that

we this is the synthetic intelligence

thing i keep you happy about instead of

artificial intelligence we know

so that's good

so that's pretty nice so that means that

what i got to do is just cultivate the

we not the i

so that jump

is what it feels like i'm wondering it

just it feels like humans could be

better served by

at least being sensitive this we're not

going to become ants

right but we would be better served by a

sensitivity of this and it feels like
they
there's a vast amount of human
de-suffering that could be made
available if if this could be made
available and i'm not sure if it's
through the arts or through economics or
what the
mechanisms of
uh influence will be to allow
the
enjoyment
of the
additional solution space or additional
um
existential space that the these
observations offer you know it doesn't
necessarily change the way we act
every day
but it definitely could
adjust the existential
aspects of the human condition and that
seems like they would indirectly serve
to reduce anxiety
and maybe hatred as a result
anyway sorry for the ramble
daniel
stephen you brought up um 300 years ago
and so it wasn't exactly 300 years ago
but it's within about five to ten years
of 300 years ago
was a book called the fable of the bees
private vices public benefits
and
i think it's it's worth reading one of
the summaries of what this poem and book
is about
um the thesis is that contemporary

society is an aggregation of
self-interested individuals necessarily
bound to one another neither by their
shared civic commitments nor their moral
rectitude but paradoxically by the
tenuous bonds of envy competition and
exploitation so it's a bit of a parable
about

private vice and public benefits in a
bee colony and about how and why that
succeeds or it doesn't

and um

let's work with the preferences and
sensory capacities and affordances that
we have

and then think about how to engineer
that with stigma g and cumulative
culture to the kind of collective
outcomes that we want to see
rather than

try to optimize it at potentially a
lower level or without the whole picture
in mind which could lead to not just um
private vice but public vice as well
nice

yeah yeah thanks um so in a way what
you're saying when he's when they're
talking about envy and such like it's
more like um

i suppose

phenotypical or social

property

rather than this

individual

oh you've got it well done so right but

now we we've tend to now focus as an
individual

behavioral

abnormality i suppose or and you lose
that capacity
or we've lost that capacity or
and hopefully now we're gaining it to
speak at that
bigger
level and not just the bigger level in
terms of a statistical social science
averaging but one where people are
actually still purposefully engaged in
the process
so that mid scale type of thing
so daniel and then i'll give my thought
oh yeah blue
oh so my um
that kind of ties back to what scott was
saying about um
you know like all the books that you
have around us and also into this like
concept of synergy and it's something
that i was thinking about like just
yesterday we had this literacy report
come out in new mexico that was just
horrible right like the kids can't read
i mean not for a long long long time but
maybe you know it made me think like we
are doing this method of communication
here like via youtube and connecting
like over support we're into this like
oratory or even like i don't know audio
visual like is there a is there a big
bigger word for that
like a video transduction of information
and it's like are we you know maybe the
need to read and write is becoming
obsolete i mean as we're moving in and
as we're
you know externalizing all of this into

our niche environment i mean
in the history of early humans history
was an oral tradition and then it became
a written tradition and now maybe the
need for reading and writing is is not
there because we're back into this
ability to directly communicate across
vast distances and to leave recorded
traces of this audio visual
communication
in time i mean yes in a media that's
less permanent i mean if the internet
survives or whatever but i mean we're
all under the assumption that you know
in 50 years if humans are still alive
this video will be archived in the
wayback machine or somewhere right and
so maybe are we are we losing the need
for that i don't know just a thought
ah damn good
so i'll give a comment on that related
to ant brains and then we can go to the
dopamine discussion
so there's um a scaling relationship
that people have observed with social
mammals
where the larger the group size of the
mammal the larger the relative brain is
so that suggests that sociality in
mammals includes like memory
the ability to simulate counterfactuals
potentially theory of mind or thinking
through other minds like we talk about
in active inference but being social for
mammals is clearly like a cognitive load
and in the ants
a trend in the other direction is
observed where larger colony sizes have

often smaller

relative brain size of the workers

and that's one of the key differences

between social and youth social so a lot

of times people will talk about the bees

and the ants as being social as if

they're hanging out or doing it because

they like each other but the reality is

that ants have been colony living for

120 million years of evolution slash one

creation however you want to look at it

and they're not doing it just to hang

out

so it shows that once you make this

phase transition from social which might

have increased cognitive burden to you

social it might be associated with

decreased cognitive burden and decreased

cognitive necessity of the individuals

because it's like it's good to follow

the trail and go off on your own and

just wander around and leave the trail

when you find new food etc those are

simple rules

that aren't necessarily requiring

tremendous onboard cognition yeah scott

maybe that's why um populations of

humans don't have to be educated

maybe we're going back to that and say

maybe we don't need high levels of

teaching for everybody in society

there will always be there's always

going to be a distribution of education

and many dimensions of that education

and um

all we can hope is that the cumulative

context supports everyone i mean the

reality is even even if we tried not

everyone's going to be a brain surgeon
and a rocket scientist and an ant
scientist and this and a lawyer and a
doctor and a grocer right so we have to
specialize and
parse and so it's kind of interesting
because the aspiration
of self
cultivation suggests that you want to
have a you know
strong body strong mind and every
individual needs to be the
representation of everything but it's
not maybe that
aspiration
of power being residing in an individual
through individual capacities is
not scalable you know even though it's
observed in mammals at a certain scale
maybe at a certain scale you have to be
used social because the intrinsic
complexity of the society and the
information available is such that it's
impossible
to reside in a single
brain
but the again
one of the things i say is that
i
tell people the brain is infinitely
capacious because it's just an antenna
and it's just tuning into the mind which
itself has whatever capacity every
capacity
so again maybe we need to train more not
in substance but in process of learning
or in knowledge discovery rather than in
facts

so that people can
navigate the mind that's collective
with their brain that's an antenna
i have a similar thought actually scott
like maybe i mean we're we're becoming
so
connected like you know in
even just 100 years ago we were
very dispersed and so maybe as we're
becoming more connected we're becoming
more like a high having developing more
of like a high
mind where like the reliance on
individual
survival or cognition is is less so
stephen and then we'll go to daniel and
ant brains i'm just going to say just in
instead of just in time manufacturing is
just in time thinking
and also tying into that process piece
um you know how much
it is about
getting the knowledge i either book off
the shelf and how much is it knowing
the policies for how to engage that book
can be the thing and that's where it can
be valuable to see an old video ought to
have met you know someone who actually
met einstein
would have known that other side to how
he
wrote those books right we just have
this encoded form
and there's other ways things can be
encoded right
um and that may be what you experience
when you have a
chat with someone maybe a professor or

someone who wrote the book which is very

um

there's there's there's that quality to

it and that that might be um

something where

thinking about because in a way these

ants really they've got to

have trust in their action policies

scaffolded by

some sort of

pheromone trail

and we in if you actually look at most

psychiatry is basically taking the

individual and trying to maybe reduce

their anxiety or the distortions

changing serotonin or

dopamine levels

um

and i think what this is showing is well

you know

that might make someone less anxious and

less

you know less challenged in their

integration into society but there's

another there might be other ways

and that speaks to eco psychology and

other things which maybe seemed a little

bit hippie before but they can be

scientifically rigorous

from what we're talking about and

that's great

so just a quick little interject

question because steven brought it up

and this is something that i've been

like thinking about and i was gonna

bring it up with with the counter

factual point but like is there a role

for trust in the social insect synergy

like like can you maybe daniel has
examples of ants that are liars or ants
that can't be trusted or like is there
some kind of like deviant form of ant
thought or behavior and then what it
brings
i'll give one interesting example of
that which is
um under the clean division of labor
model one would expect that queens
should just be like super fertile and
workers should never have any capacity
for fertility
and so when it was found empirically
in honeybees and in some ants that some
workers had partially activated ovaries
like they retained the capacity to
reproduce
which is also facilitated in those
insects because unfertilized females can
make male eggs just by making a haploid
egg
initially that was framed as cheating
and it was in the um the game theory of
cheating and all of these ways that you
could think about economic exploitation
free rider problem
but at the same time it's the case that
queens do die
and when that happens the call in me is
almost hedging its bets against one
failure mode by having this just-in-time
capacity to salvage some reproductive
fitness with pre-activated workers
so was it deceptive at one level was it
adaptive at another level there's many
examples like that and i would say it
all falls under that ambiguous stimulus

heading i'll ask a question from the
chat and then we can because it will
take us towards the brain
um sergio asks
how can the autonomy of an agent for
example an immune b cell living within
another autonomous meta agent like a
mammal
be accounted by colony behavior and
stigma g
it's an awesome
question it speaks to the nested
formalisms
so ants also have mobile cells and we're
abstracting around that when we just
have a nest mate level model
but potentially nested modeling could
speak to that because t cells um are
perceiving their local niche
and then they're they're acting in their
niche they're releasing cytokines and
they're um changing the immune context
which could bring other t cells or b
cells down and then i posted a paper
distributed adaptive search in t cells
lessons from ants with professor gordon
and others that was explicitly about
that it was like a triple play
collaboration with ants immune cells and
computational search algorithms
and the way in which computational
search algorithms and computational
frameworks like active inference
although not in that paper
give us a common descriptive ability
across
the ants and the t cells
but then we once we frame it we learn

from the ants and the t cells and the richness of nature and that expands our computational models so it's like a feedback in the informational niche between the kinds of cognitive and computational models we propose that helps us generalize and then we get surprised by ways that nature actually works out that feeds back into how we modify our models so maybe anyone else can give a

comment on the the b cell and the ants and then otherwise we can go to the dopamine

you think we're good

yeah to go to the ants in the dopamine actually

marco i was curious how i know it's a big question but how has dopamine been seen

in the human neuro community and clinical side and where does act inference

come into play or expand on that because i suspect that

we might see some parallels with what it's doing in ants as well but what is the situation been and where is it going with human dopamine

well i think that depends on who you're asking

there is for sure one big uh focus on uh the rewarding effects or let's say the role of dopamine in addiction

um and you know just as as a positive prediction error signal

um but then of course you have so and

and that
you might put at the
um you know at the dopamine uh
tract that comes from the vta and goes
basically throughout the the cortex and
also to the nucleus accumbens but then
you have also the the movement
facilitatory role of dopamine which
stems from the substantia and
innervates the striatum and
basically serves a
action policy selection role
via selecting competing affordances i
guess you could say
and and by that uh you select one um
movement uh strategy
and if uh
you lack dopamine then uh basically you
um you're freezing in your movements and
that's one of the cardinal symptoms that
you get in parkinson's disease
and as an
overarching principle i guess at least
that's the intuition i have right now is
that dopamine somehow modulates uh how
some
one like an agent interacts with the
environments it's it's more this um this
interface
how how do you weight
uh signals from the environment that you
can react to
and and that's i think the aspect that
combines the the rewarding function of
dopamine with the
motion and action um
permitting function
yeah

cool i see a lot of interesting
parallels with the ant world so
first for those who might not be
familiar
ants and invertebrates have a lot of the
same molecules in their brain as mammals
do so like some of the same exact
neurotransmitters like serotonin and
dopamine which is why some of the same
drugs that mimic the structure of those
neurotransmitters have effect in
invertebrates as well as in vertebrates
but they don't have the same
anatomy of the brain so like you
mentioned the vta and the substantia
the ants and the invertebrates
don't have those exact structures
however they do have some structures
that play similar computational or
functional roles so point one is that's
an argument for why we need to look
beyond just the brain layout of the
mammals
because not all cognitive systems
inside or even beyond animals have that
exact layout so it motivates just like
we've been talking about like looking at
the specifics really understanding how
does
these different dopaminergic signaling
pathways work in the mammals but also
what are they doing functionally and
computationally so that we can
understand and design other systems
and then um
dopamine has been studied
for playing a role in the regulation of
individual foraging

in invertebrates as well as in vertebrates so people have done experiments with giving dopaminergic drugs to insects and seeing how it increases their foraging activity and even in the ants that are um shown on like slide 14 those ants the ant going into the nest entrance there is painted because it's been treated with dopamine so it's possible to collect ants and then either treat them with dopamine and measure how much it influences their dopamine levels in the brain we found that increases in brain dopamine through pharmacology increased foraging activity and we found that providing an inhibitor of dopamine synthesis decreased ants foraging activity relative to like a control group on the same day so on first pass that is consistent with this reward role for dopamine and that's the kind of empirical observations that people have found confirms their belief that dopamine is playing a role in reward but then what you said at the end there that reward and action come together with sensitivity and so we can actually add in something that we know about these ants and about other ants to think about how does dopamine play that role well in this ant species we know that individual nest mates make the decision to go foraging or not probabilistically depending on the rate

of interactions that they have with incoming successful forager nest mates so if there's a high rate of incoming successful foragers it stimulates more foraging and if there's a low rate it sees this foraging activity it's like if you block incoming foragers from coming in there's a lag followed by a drop off in foragers leaving and if you have an excess of foragers coming in it leads to an excess going out which eventually does deplete the workforce so it's not like there's infinite ants they can just keep on sending but that's a self-regulating process so that when things are good and food is coming in send more it's a good day to forage but if there's only a trickle or there's like a predator blocking one of the paths then no more ants go out and so one of our hypotheses although it would take a few other kinds of experiments to really go into this in detail would be that dopamine plays a role in modulating the sensitivity to interactions of nest mates with each other and their sensitivity to changes in environmental conditions and so by framing what dopamine does as modulating precision or modulating sensitivity to interactions we integrate the mechanisms that we know actually underlie the regulation of foraging in ants with the empirical outcomes that increases of dopamine signaling

increased foraging activity and
decreases decreased foraging activity
uh so stephen and then marvin
i'd be interested where your thoughts
are in terms of the type of foraging
whether it's kind of an anxious
like foraging if that makes sense
because that could be because often um i
think that some of the serotonin
inhibiting uh
medications they're helping
to reduce anxiety and there could be a
difference between
the way people are reading those sensory
inputs
so i suppose could be could the ants be
also foraging more
if they're anxiously foraging if that
makes sense maybe a sudden peak and then
at some point they
they there's that that doesn't become
such a dominant dynamic for
addressing reward
do you want to respond daniel and then
we'll go to marco
yes there hasn't been a huge amount of
work on serotonin foraging but there is
one paper i'll post in chat which is
serotonin modulates worker
responsiveness to trail pheromone in the
ant *phytoleontata* so it is known that
different neurotransmitter systems play
a role in modulating the sensitivity to
interactions and whether that gets
psychologized or projected again
i don't think that's always the right
move to make but empirically serotonin
does play a role there

so that makes me wonder is um
so the the neural modulators that you
usually find in in
mammal brains do they all exist in uh in
ants as well like uh including um
norepinephrine acetylcholine etc
so it's a very interesting question
there's some that are kind of like a
level celebrities in vertebrates and
invertebrates like dopamine and
serotonin

then

epinephrine and norepinephrine which are
very important for the regulation of
arousal in vertebrates
their functional roles are to a large
extent played by actually the
neurotransmitter pair of tyramine and
octopamine in invertebrates which are
trace amines

received at the trace amine receptor in
vertebrates and so the exact proportions
of these different neurotransmitters
differ

in vertebrates and invertebrates but the
metabolic enzymes
are

in the same gene families and that was
some of the most interesting analyses
was actually to look at the sequence
evolution

and the duplication and deletion events
of the metabolic enzymes that transform
neurotransmitters into each other from
common amino acids into
neurotransmitters synthesizing and
degrading them

and then the protein receptors that are

used on the cell membrane like the dopamine receptor so the gene families are homologous the chemicals are similar

and um neuropeptides there are some that are ancient gene families that are shared across animals

and there's other vertebrate specific and invertebrate specific neuropeptide gene families so they're more similar than dissimilar

and there's a massive amount of overlap and that's why again like drugs that are designed for working in one species often have a similar effect like neonicotinoid pesticides

they affect the acetylcholine receptors of invertebrates broadly

just like nicotine affects the acetylcholine receptors of humans

right um yeah i think the the really cool um

promise of active imprints is that you can understand different cognitive systems

in terms of the messages that are being passed

and

and there are lots of comparisons you can make and

i guess that's also a good um

general test of the of the substance behind the whole framework

to um to try and

make those mappings

where you say you have

signals that get passed from bottom up

like typically

by a glutamatergic um
trains of
spikes let's say
they might be matched by expectations
that come from top down which might be
inhibitory and then you have all sorts
of uh modulatory uh signaling molecules
that
that's the precision of certain messages
there
so and and once you have that framework
established you should be able to find
it in all kinds of different cognitive
systems whether they are structured like
the brain where you have
physical pathways that link one region
and another or whether they are
let's say

[Music]

more based on
externalized
elements like uh pheromone trails or
whatever
so
i mean very much at the beginning but i
think there's lots to discover
yeah marco i just want to respond to
that really quickly you know we were i
don't know if you caught our last live
stream i think it was 28.2 just the one
that was last week but we were talking
about um
you know it's interesting we're talking
about mental wandering like what does
mental wandering have to do with like
foraging or ants but there's a lot of a
lot of connections to be drawn and
really like

we're thinking about um
you know tracing a thought through the
brain like like which is i mean my
background is also neuroscience which is
like passing a message from one neuron
to the next neuron and really the basis
of the paper was mental wandering and
meditation and like the you know not
suppression of thought but like when you
consciously inhibit
you know the thoughts from wandering
around so it says mindfulness practice
and
it's really interesting because in the
ant colony there's so many like
relationships that i can see like i mean
even between like foraging like when
you're wandering and you're mental
wandering is that like thought foraging
or but i mean also more than that
because there
is that like synaptic release and so is
that i wonder if that like synaptic
release or synaptic transmission that
happens in neurons is similar to this
like pheromone deposition right so like
you have synaptic release and that
encourages more neurons to speak
signaling so you have this pheromone
deposition and that encourages more ants
to keep foraging on those trails and so
i really do wonder
how much of the what our role
is in neurotransmitters is externalized
in this kind of niche
modification and external cognition
which would happen for us internally
like but i wonder if the answer are

doing that as part of this like high
mind mentality

um sorry steven i i was rambling yeah i
know it's very very interesting points
one thing i really picked up as well and
i um with what you're saying in marco is
this message passing that you have in
active inference it is very liberating
and
and

because you traditionally talked about
in terms of signals and signals ends up
maybe being electrical signals and it
ends up being like a wire that we think
of in the you know other
like electronic world and and it's a
very big trap you know so then there's
somehow a signal going through and then
next thing you know you you know
everything implicitly or explicitly
becomes a computer
but this this just this general idea of
message passing somehow
so signals still it's hard not to make
signals about you know a transmission
system more like in a
in a wire but message passing it at
first i used to wonder why they're using
that term because it seems so ambiguous
and like what's going on with that but
in some ways i can see is so much
benefit from that
from that that formulism
cool daniel
yes the signal
and signal transduction signal
processing which is frameworks that are
used in electronics but also we talk

about like signal transduction in the
cell so it
shows how wide the usage of that signal
term is
and it almost implies an essentialism to
the meaning of the signal like it
actually carries a meaning but when we
think about message passing we can be
clear about what that message is
and then
look at how the generative model
interprets
what is the incoming stimuli or
observation
instead of treating the signal as what
is bearing meaning we can think about
the stimuli as just being the empirical
observation which as we've been
discussing from the sensory level on up
through the narrative
observations are ambiguous
and they require a generative model in
order for sense making to occur and so
agreed that not only is message passing
away that bayesian models can be updated
like we learned about from bert devries
and others but it's also a way that we
can step away from a sort of signal
versus noise partitioning paradigm
and move into an empirical observation
meeting
generative model meaning maker
marco yeah uh so what i also like about
this but i'm i might um have a
completely different perspective on this
than than uh
you know you have but um the the
the nice thing about messages is that uh

it takes you away from
this object orientation where you
identify things and you can say they
either are okay or they're
malfunctioning
but you rather look at the interaction
between
let's say vertices or objects
and and you find
maybe more illuminating answers there in
if you think about
psychopathology for example
so that you can point to one brain
structure and you say ah that brain
structure is broken and that's why i
have symptoms it's rather
uh
to think about it in terms of
well maybe
this message cannot be transmitted
and uh and that's uh maybe the reason or
or that's where you have to look for
answers
uh so one other thing about message
passing that's i think a great way to
think about active inferences it really
grounds it in information theory because
that's like shannon information is is
like you know spawned from that idea of
message passing and coherence and a
message but also in the active inference
framework it's always subject to
interpretation
uh stephen
yes i um i was as i started to think as
you were saying that marco about
someone shaking hands with someone else
okay so

you transmit a handshake
you pass a handshake you know i mean
because both people have to reach out
whereas when you transmit a signal it
can be a one way a more than one way
thing i suppose in some ways i think
i i sort of agree what you're saying
blue i also would say in some ways that
shannon entropy is probably a bit
stronger on the signal side
um

so in some ways it it gives that
effective message passing but active
inference goes further down well it
seems to offer further down that route
because

it gives this um
other way to extract so it's not like
the noise is diminishing
the signal um although
it's it's there's other ways of getting
that through action
um what's in the signal um so i suppose
in a way it's a bit like the handshake
it's like what's that hand saying to me
the funny handshake from the mason well
i reach out and i hold it you know
and i get a bit of a better feel and i
see what i want to pass up my uh
predictive processing chain
daniel

it just makes me
think about how when we think about
signal occurring amidst noise it's
almost like we're getting one unit of
stimuli whether it's rgb channels on the
tv or whether it's a frequency range for
audio or whatever it happens to be

if the task is to
find the signal in the noise we're
always going to get less than one could
be 99 signal to noise it could be 5
so we're kind of always going less
when we think about
the stimuli as being a message that's
passed to us and something that we
enrich
we can make it more than one so that
secret handshake or that shape
can be
unpacked and understood in a broader
context so for example how many you know
bits of information does it take to
communicate how to make lasagna or
something it's like you don't need to
describe every single
sequence
somebody who knows it only takes one bit
of information maybe just the letter I
or something like that
and so
signal processing again makes us think
about diminishing and filtering out
what we're getting in
whereas maybe active inference helps
moves us towards understandings where
there's an enrichment of the information
coming in which is always required but
especially when there's sparse or
ambiguous sensory stimuli
nice steven
that's a good point daniel i mean that
sort of takes you to the value of varied
stat fairy tales and yeah the idea i
hadn't really thought about that before
but the idea that

the generative model
is free to
make more of something um and embellish
it i suppose
and of course that that may
also really come into its own when
you're bringing in multi-sensory
information so you know
there's there's this uncertainty but
there may have to be something which
embellishes
um early on maybe that explains why
children early on
get very into dreams and uh
very wacky sort of modeling of the world
because
um they maybe want to extrapolate beyond
you know what that sound and that light
source and that feeling and that smell
of food is you know the porridge for the
three bears that goldilocks is eating is
is actually going to signify for them so
yeah that's interesting
so
did we already have this like
conversation about do ants dream of
undead crumbs did we have that
conversation daniel is there a book or
something right
i think you dreamed it okay okay so that
brings me to my next question was
it you know i wonder like i i just was
um listening to josh steinbaum's lecture
and i read his dream code dreamcoater
paper in which like the computer
explores like potential recombinations
of symbols and stuff in like a it's like
an off phase like a helmholtz machine

there's like a wake sleep cycle in this algorithm that um he was using and so you know i mean i wonder i guess ants sleep i mean they like rest do they rest they have to like assimilate and update their generative model do they believe so

people have studied circadian rhythm and the foragers have a strong circadian rhythm again probably not every single species but we did some work with gene expression and found gene expression cycling of circadian rhythm genes in the brains of foragers but not nurses in this species so nurses always indoors low-key working all the time they take a lot of micro naps and foragers have more like phasic circadian rhythms especially for insects that use the sun to navigate like honeybees well so what i really wonder is what about the collective um like ant

generative model right like you think about the the this is like a collective behavioral or collective consciousness like is there you know dreaming or potential like recombination like exploration does that happen in like an ant colony

it's an awesome question and another example of um true colony phenotypes and people have studied the activity waves in ant colonies

over multiple time scales like a minute time scales as well as the circadian rhythms and that is actually not dissimilar from the multiple bands in

the eeg

so just like the electrical dynamics of
the brain which present as one sort of
just time series

you can do a decomposition

to extract out like the alpha and the
beta and the theta bandwidths so just
like there's only you know one 2.4

gigahertz

wi-fi wavelength

yet there's multiple wi-fi networks that
can co-exist so how does that multiple
coding happen

for the signals that we make like wi-fi
and radio

for the brain which is uh using
different frequency bands simultaneously
and there's also these oscillation
dynamics in ant colonies

so if the whole colony and this a wave
like what i mean what kind of signal
technology are you using to detect these
waves i have to

i'm not

um

similar to the way that the time series
of the eeg

you would do a sort of uh four-year
decomposition to look at the different
frequency modes

similarly people look at the spatial
position of ants and then can ask what
are the activity dynamics through time
and one paper i'll post in the chat is
from 2016

by gel

all emergent oscillations assist
obstacle negotiation during ant

cooperative transport
so actually having this sort of like
oscillating
group activity
helps them not lock on
too specifically to an object's
direction that it's being pulled
like if there's some object and then
there's the nest entrance yeah okay
we'll pull in a straight line isn't that
the shortest distance between two points
yes but first off no one knows exactly
where the nest entrance is
and second off even if it were the most
direct it might not be the most
resilient or tractable
but if you just say okay you know we're
gonna pull a little bit this way and
zigzag it this way and zigzag it that
way you get that wisdom of the nest
mates because they each have a different
estimate on exactly which way to go they
get averaged out and if there's a little
block somewhere
built into the movement that this object
is being brought home with is the
ability to go around it
and so that's another example about how
processes that are dynamic at the very
very micro level with the agents
often constitute more adaptive emergent
processes and more tractable ones it's
like we either get the resilient
tractable agent level rules or
we get the fantasy well we should just
move it in a straight line everybody
should know exactly where to go and if
they don't we're going to punish them

it's not how it works
awesome um and so that kind of leads
nicely into i think we were talking
already actually about like the wisdom
of the crowd um and the example with the
ox maybe that was in the dot zero or
maybe last week but um
it really leads me to thinking about um
collective behavior and
collective intelligence and just um that
might be like a nice
question for people to give thoughts on
because i i just it's in my own interest
to find out what other people think
about the difference between collective
behavior and collective intelligence and
where is
um you know any thoughts there on that
stephen
yeah
sort of linking that and what you're
just saying now with what daniel was
just saying
in some ways the brain so if you've got
going more chaotic which i suppose in
some ways you can do through your
environment or just
and the environment will do it to you
um
and then you've got this synchronization
of
waves
as was mentioned
so you could go from that kind of
synchronization of waves and i suppose
it could become more coordinated
or it becomes more chaotic so you've got
this kind of balance in terms of sense

making between
more entrainment between
more coordination of action either
between agents or between the
sensory motor
properties of the organism
or
which would in some way
be more focused you would imagine
and has that risk of rigidity but has
the advantage of efficiency
or
um so brainwaves seems to have a bit of
a bit of a mid you know this this
synchronization of brainwaves gives you
a sort of a mid-scale potential so it's
not completely chaotic but it's not
completely
you know synchronized
nice
yeah i don't know i'm not sure
what
the difference is really between
collective intelligence and collective
behavior
i um
there's this cool diagram collective
behavior and collective intelligence
and it shows cooperation and
coordination as essential to collective
intelligence but but it also then like
that those are both behavior cooperation
and coordination and then what's left of
cognition
and so
it's cognition intelligence or
like what is cognition anybody have any
like thoughts on i'm not marco i know

your uh background is neuroscience so
let's let daniel think and then i'm
going to put you on the spot
studying
collective systems
it always struck me as interesting
because the organism is a collective of
cells and so it's more like an approach
that we take to any system to see it as
arising from interactions at an even
lower level
and so when i look at these very human
centric
um
thought maps informal thought maps
it includes things that you know open
source software p2p business
those are human-centric and so it ends
up
muddying the waters and introducing a
lot of psychology and game theory
and terms that aren't like distinct like
is cooperation distinct from a
coordination
and that's one reason why having a
perception cognition action model
even though it comes across as very
austere
and distant from the phenomena that we
want to explain like coordination
cooperation and cognition
having such an austere but scalable and
flexible model
helps us get at these terms
without presupposing them as natural
things that exist in the world so we get
to just model what is on the ground
happening like with the ants or on the

ground in another context and then ask
how our definition or parameterization
or formalization
of coordination and cooperation are
coexisting as descriptions
of the process that we're specifying at
the actual proximate mechanism level
nice stephen and then marco
yeah i think that
that challenge of these things having a
psychological or
folk psychological or cultural
labeling
is limiting and
by stepping back from that
might which in some ways makes creates
more uncertainty i could say that it
seems more austere but it's more austere
because it is less
clear initially but the benefit of being
less clear
and this actually ties in a bit with
some of the infusion space work that
i've been working on is it makes it more
nascent it um whilst with active
inference it gives that holding
container so by being more nascent
you've got more ability to then proceed
again with another type of naturalistic
potentially more naturalistic framework
and not being tied to a naturalistic
framework
based on language and all the
distortions that can come from that so i
think that that's a really big part of
what makes it useful
oh sorry
no thanks um

these are all super good points um and i agree uh completely and i think that this is also recognized in um in psychology that there is a tendency to attribute all kinds of uh terms to uh different brain areas for example um like the prefrontal cortex there are more functions assigned to that it could possibly hold for example you could find papers that identify the location of um stock trading behavior and these are clearly human invented terms and there is no reason why they should have um an expression in the in the brain um and so um i think uh a nicer way of thinking about this and um i think you mentioned it before is uh this dynamic between segregation and integration so sometimes it makes sense to be focused on a particular task everyone let's say uh you know without in lots of distractions you you are very efficient at performing one particular task um but that might not be the most creative mode of operation um where you might rather want to have lots of different brain regions

simultaneously interacting with each other and exchanging lots of information that's not efficient but it's very integrative

so um neither of those extremes is probably viable by themselves but if you have a good

flexible way of moving between segregation and integration you can be very much task oriented specific but you can also be creative

and incorporate lots of different information

awesome daniel

it's a good point marco and also that meta task of appropriately choosing how to shift tasks

is framed as a type of mental action in the nested active inference the deep active inference formalism

so we can actually think about that shifting

between different

affordances at the attentional level

as another kind of action selection task that uses this same perception cognition and action framework so and i agree as well stephen it does

start with a more

atomic germ

it's more nascent

and that's what allows us to map and bridge and compose

if we try to put too much meaning into the lowest level

we actually constrain how tall the buildings that we can build are

and douglas hofstadter talks about this

in the context of reflexive computer
systems

brains and ant colonies

just today as it was in 1979 three of
the most important systems to understand
and gets at in all of those systems how
the lowest level

like the interactions among nest mates
themselves or the letters they cannot
have meaning in the way that for example
words do

because if you load all the meaning of
words onto phonemes

you can't compose sentences

and it's just it's very interesting how
even though of course we want that
meaning

we have to actually take a dip into
using

pieces that are sub meaningful

in order to have flexible meaning

because if you just try to run straight
for the meaning you're going to end up
with a very constrained system that
might have the one function or meaning
that you want to encode but it won't
have the ability to reflect or to
generate novel meaning

nice

so just in the last few minutes um i

just want to thank daniel for taking the
time out to discuss his awesome paper
with us which i've been really wanting
to discuss for several months um and
hopefully he'll be back next week with
more conversations about active infrared
um and marco welcome it was really nice
to have you here uh and so just if

anybody wants to give a final thought on
this paper or some stuff that they're
excited to talk about next week
well thanks a lot this was really
awesome

and i learned a lot there are also lots
of references i think that i need to
check out now that you mentioned
um and yeah really thanks a lot for for
the paper and for presenting today
yeah just to mirror what marco says and
also welcome marco thanks for joining us
um

really really enjoyed the the different
avenues

and actually the ability for it to
keep coming back to something which
could hold those different
um

branches which is quite nice so
we saw an ability to come back to
and reintegrate a lot of different ideas
so that was really nice so let's look
forward to keeping that going
cool daniel

well thanks everyone for joining and
blue for excellent facilitator and
broadcaster role

i think for 29.2 what might be awesome
other than having some novel nest mates
on the stream and in the live chat
what might be cool would be like to lay
out some of these analogies that we've
been exploring like okay agent and
environment all right that's nest mate
and pheromone density there's like the
environment that you can control and
then there's sort of the abiotic

environment or there's the parts of the environment that are outside of the affordance of niche modification like the thunderstorm okay so maybe having that kind of a table would help us align a lot of the systems that we've been talking about so ants computers brains societies

other systems that people care about that's kind of like us preemptively foraging maybe laying down the inkling of some trails because just like a few years ago when i was deep in the ant game

seeing that there was a way to bridge active inference and ant research in potentially a new way maybe someone's going to go oh i mean architecture i do that every day and i've been looking for a way to inroad active inference or another another way that maybe we could help scaffold

somebody else's research because i think the stigmatic modeling and a lot of these discussions

go to a lot of systems and things that people are already working on so that would be awesome to talk about in 29.2 and to build momentum going forward awesome well thank you again everyone and we were are looking forward to our discussion next week so get in touch if you are watching on the live stream and want to participate

bye